Exercise

Goal is to classify resturant reviews as positive or negative using neural networks. Both a feedforward neural network and a recurrent neural network are being implemented using keras

Model Architectures:

Feedforward model:

Embedding layer
Flatten
Dense layers

The model consists of an embedded layer with output dimension of 32, then a flattening layer followed by one Dense layer with 128 units using "relu" and one Dense layer with 64 units using 'relu'. Last is the output layer which uses 'sigmoid' This achieved an accuracy of 90%.

Recurrent Network

Embedding layer
LSTM
Dense layer

The model have an embedded layer with output dimension of 32, then a LSTM layer with 64 units and last a Dense output layer using "sigmoid". This achieved an accuracy of 82,3%

```
if model_type == "feedforward":
    model.add(keras.layers.Embedding(input_dim=input_dim, output_dim=32))
    model.add(keras.layers.Flatten())
    model.add(keras.layers.Dense(128, activation="relu"))
    model.add(keras.layers.Dense(64, activation="relu"))
```

The feedforward model

```
elif model_type == "recurrent":
    model.add(keras.layers.Embedding(input_dim=input_dim, output_dim=32))
    model.add(keras.layers.LSTM(64))
    model.add(keras.layers.Dense(1, activation="sigmoid"))
```

The recurrent model

```
model.fit(
    x_train,
    y_train,
    epochs=2,
    batch_size=64,
    validation_split=0.2,
    verbose=1
)
```

The fit method, training for 2 epochs and a batch size of 64

```
model.compile(
    optimizer="adam",
    loss="binary_crossentropy",
    metrics=["accuracy"]
)
```

Compile method with "adam" as optimizer

```
(venv) simenaure@simen-macbook Assignment 10 % python3 assignment_10.py
/Users/simenaure/Desktop/tdt4171/venv/lib/python3.9/site-packages/urllib3/__init__.py:35: NotOpenSSLWarning: urllib3 v2 only supports OpenSSL 1.1.1
+, currently the 'ssl' module is compiled with 'LibreSSL 2.8.3'. See: https://github.com/urllib3/urllib3/issues/3020
  warnings.warn(
1. Loading data...
2. Preprocessing data...
3. Training feedforward neural network...
Epoch 1/2
4914/4914 ━━━━━━━━━━━━━━━━━━━━ 6s 1ms/step - accuracy: 0.8626 - loss: 0.3072 - val_accuracy: 0.8953 - val_loss: 0.2433
Epoch 2/2
4914/4914 ━━━━━━━━━━━━━━━━━━━━ 5s 1ms/step - accuracy: 0.9024 - loss: 0.2302 - val_accuracy: 0.9017 - val_loss: 0.2305
4879/4879 ━━━━━━━━━━━━━━━━━━━━ 1s 346us/step - accuracy: 0.8993 - loss: 0.2378
Model: Feedforward NN.
Test accuracy: 0.900
4. Training recurrent neural network...
Epoch 1/2
4914/4914 ━━━━━━━━━━━━━━━━━━━━ 44s 9ms/step - accuracy: 0.8672 - loss: 0.3023 - val_accuracy: 0.9160 - val_loss: 0.2023
Epoch 2/2
4914/4914 ━━━━━━━━━━━━━━━━━━━━ 44s 9ms/step - accuracy: 0.9204 - loss: 0.1949 - val_accuracy: 0.9257 - val_loss: 0.1790
4879/4879 ━━━━━━━━━━━━━━━━━━━━ 9s 2ms/step - accuracy: 0.9233 - loss: 0.1856
Model: Recurrent NN.
Test accuracy: 0.923
```

The achieved results, 90% for feedforward and 92,3% for recurrent
The recurrent model performs better due to it captures sequense information in the text,
while the feedforward model does not. One side effect is that the recurrent model is slower to train